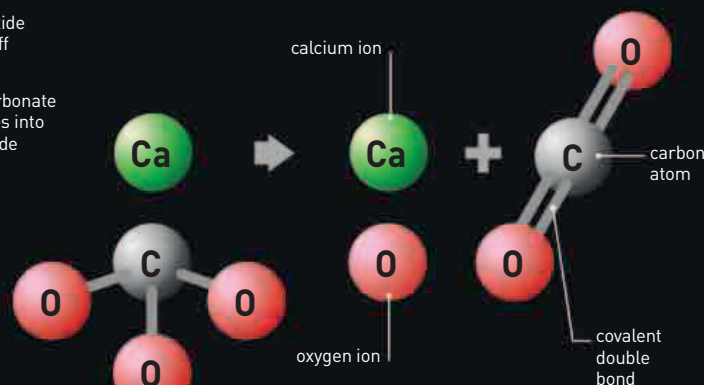
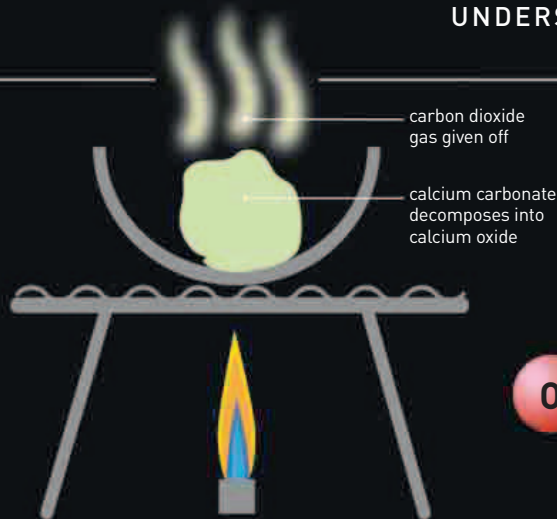


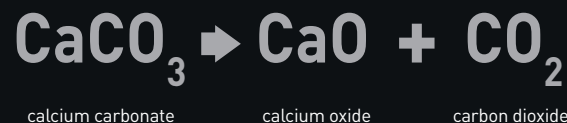
## TYPES OF REACTION

There are many different types of reaction, including, for example, electrolysis (in which an electric current splits a compound into its constituent parts) and acid-base reactions (in which an acid and base, or alkali, react together). In general, however, reactions can be categorized into three main types according to what happens to the substances involved: decomposition, synthesis, and displacement (or replacement) reactions. In a decomposition reaction, a compound breaks up into smaller parts. Synthesis is the opposite: two or more compounds combine to form a single product. In a displacement reaction, part of one compound breaks away and becomes part of another.



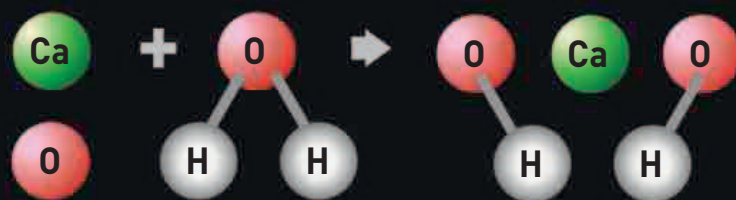
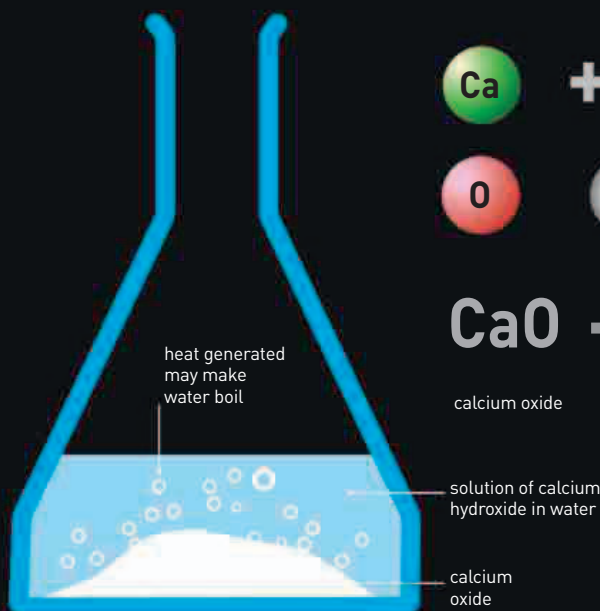
### DECOMPOSITION REACTION

Heating the mineral limestone (calcium carbonate) causes it to decompose into calcium, oxygen, and carbon dioxide. The calcium and oxygen are in the form of ions (charged particles); they form the ionic solid calcium oxide. The carbon dioxide is a gas composed of covalently bonded molecules.



“THE WORLD OF **CHEMICAL REACTIONS** IS LIKE A STAGE, ON WHICH SCENE AFTER SCENE IS CEASELESSLY PLAYED. THE ACTORS ON IT ARE THE **ELEMENTS**.”

Clemens Alexander Winkler,  
German chemist, 1887



calcium oxide

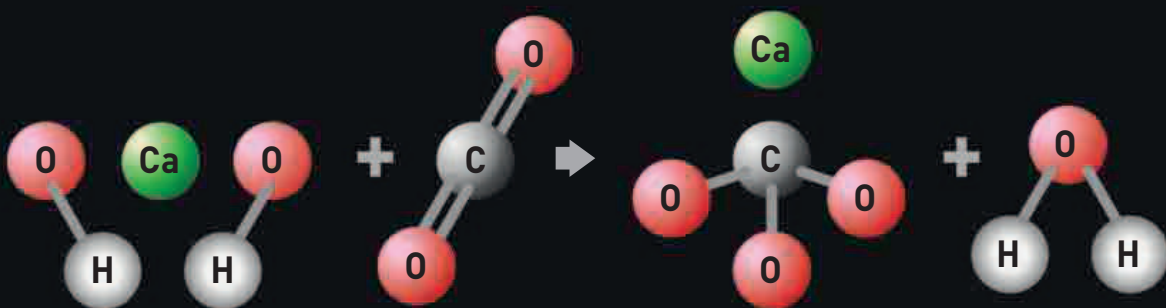
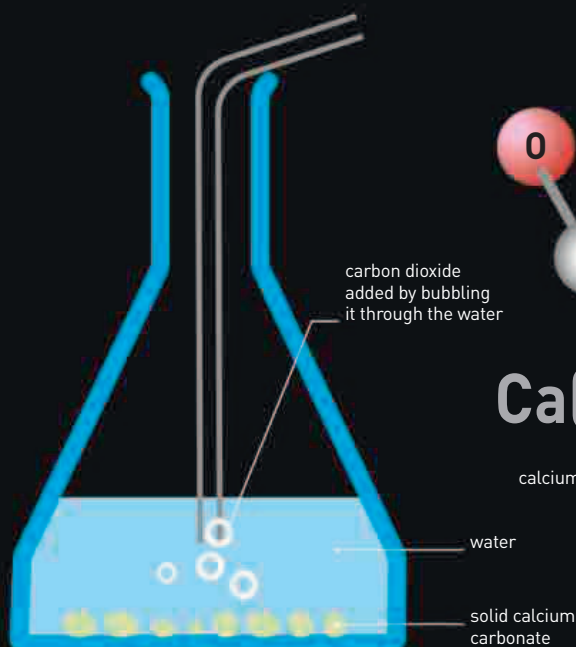
water

calcium hydroxide

brackets show that there are two hydroxides (OH) for every calcium atom

### SYNTHESIS REACTION

Adding calcium oxide to water produces calcium hydroxide, which dissolves in the remaining water. This is a synthesis reaction because the product is composed of all the atoms of the two ingredients.



calcium hydroxide

carbon dioxide

calcium carbonate

water

### DISPLACEMENT REACTION

When carbon dioxide gas is introduced into a solution of calcium hydroxide, the carbon dioxide displaces the hydroxide in the calcium hydroxide, producing calcium carbonate and water. The calcium carbonate forms solid particles in the solution.